

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 2 – Case Study
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	B. Tech AI&ML (2024-2028)	Date:	12/08/2026

Code of Conduct

I T. Bhanu Pavan Varma(192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$
- $C(\text{science is}) = 2$
- $C(\text{science drives}) = 1$
- $C(\text{science drives innovation}) = 1$

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- $P(\text{is}) = 0.66$
- $P(\text{drives}) = 0.33$

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability $P(\text{science} | \text{data})$. Interpret how this probability influences next-word prediction in a mobile keyboard application.
 2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
 3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
 4. Calculate the Entropy of the distribution $P(\text{is})=0.66$ and $P(\text{drives})=0.33$. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.
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CASE STUDY 1 – Smart Mobile Keyboard Prediction System

The system uses the corpus:

“Data science is powerful, data science drives innovation, data science is evolving.”

It applies Bigram, Trigram, Backoff, Deleted Interpolation, and Entropy analysis.

Q1. Calculate MLE of $P(\text{science}|\text{data})$

Given:

$C(\text{data})=3$ $C(\text{data},\text{science})=3$

For a Bigram model:

$P(w_i | w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$

Therefore,

$P(\text{science}|\text{data}) = \frac{C(\text{data},\text{science})}{C(\text{data})} = \frac{3}{3} = 1$

Answer:

$P(\text{science}|\text{data})=1.0$

This means that **100% of the occurrences of "data" are followed by "science"** in the given corpus.

Therefore, when a user types **"data"**, the keyboard can confidently predict **"science"** as the next word.

This high probability makes the prediction highly reliable for the given training corpus.

Q2. Backoff for "data science improves"

The sequence **"data science improves"** is unseen in the training corpus.

The trigram:

$P(\text{improves}|\text{data},\text{science})$

is therefore:

0

The Backoff model next checks the lower-order Bigram:

$P(\text{improves}|\text{science})$

But **"science improves"** does not occur in the corpus, so this probability is also:

0

Finally, it checks the Unigram probability:

$P(\text{improves})$

The word **"improves"** does not occur in the given corpus, so:

$$P(\text{improves})=0$$

Therefore:

$$P(\text{improves}|\text{data},\text{science})=0$$

Analysis

Backoff is important because it allows a system to use lower-order information when a higher-order N-gram is unavailable. However, in this particular case, even the unigram **"improves"** is absent.

Therefore, ordinary backoff cannot produce a non-zero probability. A smoothing technique would be required to handle such an unseen word effectively. This is important in real-time keyboards because users constantly type new words and combinations.

Q3. Deleted Interpolation for "data science is"

Given:

$$\lambda_1 = 0.5 \quad \lambda_2 = 0.3 \quad \lambda_3 = 0.2$$

where:

- λ_1 = Trigram
- λ_2 = Bigram
- λ_3 = Unigram.

The interpolation formula is:

$$P(w) = \lambda_1 P_{tri} + \lambda_2 P_{bi} + \lambda_3 P_{uni}$$

Trigram probability

The corpus contains:

data science is powerful
data science drives innovation
data science is evolving

Therefore:

$$C(\text{data},\text{science})=3$$

and

$$C(\text{data},\text{science},\text{is})=2$$

Hence:

$$P_{tri}(\text{is}|\text{data},\text{science}) = \frac{2}{32}$$

Bigram probability

Given:

$C(\text{science}, \text{is}) = 2$
and

$C(\text{science}) = 3$
Therefore:

$P_{bi}(\text{is}|\text{science}) = \frac{2}{3}$

Unigram probability

There are 15 total words in the corpus and **"is" occurs twice**.

$P_{uni}(\text{is}) = \frac{2}{15}$
Now:

$P = 0.5(\frac{2}{3}) + 0.3(\frac{2}{3}) + 0.2(\frac{2}{15}) = 0.3333 + 0.2000 + 0.0267$ $P(\text{data science is}) \approx 0.56$

Analysis

Deleted Interpolation combines information from different N-gram levels. The trigram provides strong contextual information, the bigram provides broader context, and the unigram provides general word frequency. Therefore, interpolation provides a balance between **reliability and coverage**, especially when higher-order N-grams are sparse.

Q4. Entropy of $P(\text{is}) = 0.66$ and $P(\text{drives}) = 0.33$

Entropy is:

$H = -\sum P(x) \log_2 P(x)$
Therefore:

$H = -(0.66 \log_2 (0.66) + 0.33 \log_2 (0.33))$ $H \approx 0.923$

Answer:

$H \approx 0.92$ bits

Analysis

The entropy is relatively low because the model strongly favors **"is"** with probability 0.66 over **"drives"** with probability 0.33. Lower entropy indicates greater predictability and therefore higher prediction confidence. A keyboard can use this information to determine whether the next-word prediction is reliable. Higher entropy would indicate that several possible words are similarly likely, making prediction more uncertain.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- $P(\text{book} \mid \text{VB}) = 0.6$
- $P(\text{book} \mid \text{NN}) = 0.4$
- $P(\text{Start} \rightarrow \text{VB}) = 0.5$
- $P(\text{Start} \rightarrow \text{NN}) = 0.5$

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
 2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
 3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
 4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.
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CASE STUDY 2 – AI-Powered Customer Support Chatbot

The chatbot processes:

1. **"Book a flight ticket now."**
2. **"This book is interesting."**

The important ambiguity is the word **"book"**, which can function as either a verb or noun. The case provides HMM emission and transition probabilities.

Q1. Assign POS Tags

Sentence 1

"Book a flight ticket now."

Word	POS Tag	Reason
Book	VB	Action/command
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Therefore:

Book/VB a/DT flight/NN ticket/NN now/RB

Here, **"Book" is a verb** because the sentence is an imperative instruction telling the user to book something.

Sentence 2

"This book is interesting."

Word	POS Tag	Reason
This	DT	Determiner

book	NN	Noun
is	VBZ	Verb
interesti ng	JJ	Adjective

Therefore:

This/DT book/NN is/VBZ interesting/JJ

Here, **"book" is a noun** because it is the subject of the sentence and is preceded by **"This"**.

Analysis

This demonstrates **contextual ambiguity**. The same word can receive different POS tags depending on its surrounding words and sentence structure.

Q2. HMM Probability for "book" as Verb

Given:

$$P(\text{book}|\text{VB})=0.6$$

and

$$P(\text{Start} \rightarrow \text{VB})=0.5$$

The HMM probability is:

$$P(\text{VB}, \text{book})=P(\text{VB}|\text{Start}) \times P(\text{book}|\text{VB}) = 0.5 \times 0.6 = 0.30$$

For comparison, for the noun:

$$P(\text{NN}, \text{book})=P(\text{Start} \rightarrow \text{NN}) \times P(\text{book}|\text{NN}) = 0.5 \times 0.4 = 0.20$$

Therefore:

$$0.30 > 0.20$$

Analysis

The HMM favors **VB** because the combined probability of the transition and emission is higher:

$$\text{VB}=0.30, \text{NN}=0.20$$

This agrees with the grammatical context of **"Book a flight..."**, where "Book" is an action.

Q3. Rule-Based vs Stochastic HMM Tagging

Rule-Based Tagging

A rule-based tagger uses predefined grammatical rules and dictionaries. For example, it can identify **"book" as VB** when it occurs at the beginning of an imperative sentence.

Advantages:

- Simple to understand.
- Easy to interpret.
- Does not require training data.

Limitations:

- Requires many manually created rules.
- Difficult to handle complex language.
- May fail with unseen contexts.

Stochastic HMM Tagging

An HMM uses probability distributions learned from training data. It considers both **emission probabilities** and **transition probabilities**.

Advantages:

- Handles ambiguity probabilistically.
- Learns patterns from data.
- Better suited for large and varied datasets.

Recommendation

For a large-scale customer-support chatbot, **HMM/stochastic tagging is more suitable** because chatbot messages can contain ambiguous words and many different sentence structures. Probabilistic tagging can use contextual information rather than depending entirely on manually written rules.

Q4. Importance of Standardized POS Tagsets

A standardized tagset such as the **Penn Treebank tagset** gives the NLP system a consistent representation of grammatical categories.

For example:

VB → Verb
NN → Noun
JJ → Adjective
RB → Adverb
DT → Determiner

This consistency helps the chatbot understand the grammatical role of words. Correct word classes improve **intent detection**, because the system can distinguish commands from objects or descriptions. They also improve response generation by providing syntactic information to downstream NLP components.

For example:

Book a flight.

The system can identify **Book/VB** as an action and interpret the user's intent as a **flight-booking request**.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
 - economic (120 occurrences)
 - growth (450 occurrences)
 - increases (210 occurrences)
 - employment (380 occurrences)

Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.
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CASE STUDY 3 – News Analytics and POS Tag Correction System

The sentence is:

"Economic growth increases employment."

Initial tags are:

economic/JJ
growth/NN
increases/NNS
employment/NN

The learned rule is:

Change NNS to VBZ if the preceding word is tagged as NN.

Q1. Apply the Transformation Rule

Initial sequence:

economic/JJ
growth/NN
increases/NNS
employment/NN

The word **"increases"** has the tag NNS.

Its preceding word is:

growth/NN

According to the transformation rule:

NNS→VBZ

Therefore:

economic/JJ
growth/NN
increases/VBZ
employment/NN

Final Answer

economic/JJ growth/NN increases/VBZ employment/NN

Analysis

The correction is linguistically appropriate because "**growth**" is a singular noun acting as the subject. Therefore, it requires a singular present-tense verb, "**increases**", which should receive the **VBZ** tag.

Q2. Analyze the Initial Tagging Errors

The initial tags are:

economic/JJ
growth/NN
increases/NNS
employment/NN

The main error is:

increases/NNS

The tag **NNS** represents a plural noun, but "**increases**" is functioning as a verb in this sentence.

The corrected tag is:

increases/VBZ

The grammatical structure is:

Economic → adjective
growth → singular noun/subject
increases → singular present-tense verb
employment → noun/object

Thus:

economic/JJ growth/NN increases/VBZ employment/NN

The transformation rule successfully uses the context of the preceding **NN** to identify the correct grammatical role.

Q3. Word Frequency Distribution

Given:

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Step 1: Total Frequency

$$120+450+210+380=1160$$

Therefore:

$$\text{Total}=1160$$

Step 2: Frequency Distribution

Word	Frequency	Percentage
economic	120	10.34%
growth	450	38.79%
increases	210	18.10%
employment	380	32.76%
Total	1160	100%

For example:

$$P(\text{growth})=\frac{450}{1160}\approx 0.388$$

and:

$$P(\text{employment})=\frac{380}{1160}\approx 0.328$$

Analysis

The frequency distribution shows that "**growth**" is the most frequent word, while "**economic**" is the least frequent among the four words. Corpus frequencies can be used to estimate probabilities in statistical NLP models. Frequently occurring words provide stronger statistical evidence, while rare words may lead to greater uncertainty. Therefore, corpus statistics can support probabilistic POS tagging and language modeling.

Q4. Transformation-Based Tagging and Entropy

Transformation-Based Tagging, such as the **Brill Tagger**, starts with an initial POS tagging and then applies learned transformation rules to correct errors.

In this case:

Before:

increases/NNS

Rule:

NNS → VBZ when previous word is NN

After:

increases/VBZ

Therefore, the tagging accuracy improves because the incorrect tag is corrected using grammatical context.

Entropy Analysis

Entropy can be used to measure uncertainty in POS assignments:

$$H = -\sum P(\text{tag}) \log_2 P(\text{tag})$$

Before applying the rule, the system may be uncertain whether "**increases**" is functioning as a noun or verb. After examining its context and applying the transformation rule, the correct **VBZ** tag becomes more strongly supported.

Therefore, the uncertainty is expected to decrease after successful rule application.

Conclusion

Transformation-Based Tagging improves the initial tagging stage by applying contextual corrections. It is particularly useful when individual words are ambiguous and their correct POS category can be determined from neighboring words. Entropy provides a quantitative way to describe the uncertainty of the tag assignment before and after correction.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____